MAHENDERKER SWAYAM KUMAR

mahendrakarswayam07@gmail.com | +91 8688556727 | LinkedIn | GitHub

CAREER OBJECTIVE

Third-year B.Tech CSE (Data Science) student with strong analytical and communication skills and hands-on experience in Python development through an internship at Coretech. Skilled in Java, Python, SQL and front-end web development (HTML, CSS, JavaScript), with a foundation in data science. Seeking an internship or entry-level opportunity to build data-driven applications.

EDUCATION

Bachelor of Technology, Computer Science and Engineering (Data Science)

Mohan Babu University

Intermediate

Sri Chaitanya Junior College

EXPERIENCE

Python Developer Intern | Coretech

- Worked on Python-based tasks using core concepts such as data structures, functions, file handling and object-oriented programming.
- Wrote clean, modular and well-documented code to automate tasks and process data.
- Gained hands-on exposure to real-world workflows, debugging and testing, and collaborated with mentors and peers.

PROJECTS

Digital Library System (MBU Digital Library)

React, *Vite*, *Tailwind CSS*, *Node.js*, *Express*, *PostgreSQL*, *Prisma*

- Built a full-stack student portal for Mohan Babu University with a React frontend and an Express REST API backed by PostgreSQL.
- Implemented role-based access for administrators and students, using JWT sessions in HttpOnly cookies, bcrypt password hashing and owner-scoped authorization.
- Developed modules for library catalog and circulation, subjects and study resources, coding practice, and a private resume and certificate area with PDF uploads.

Book Data Analytics Pipeline

Python, *Requests*, *BeautifulSoup*, *Pandas*, *NumPy*, *Matplotlib*

- Scraped book titles, prices and ratings from a website with Requests and BeautifulSoup, and exported the data to CSV using Pandas.
- Performed exploratory data analysis: checked dataset shape, missing values and duplicates, cleaned the data and ran statistical analysis with Pandas and NumPy.
- Created bar, histogram, pie, box and line charts with Matplotlib to show rating and price distributions.

IoT-Based Air Quality Monitoring System

ESP32/NodeMCU, *MQ-135 sensor*, *IoT platform*

- Designed an air quality monitoring system using an ESP32/NodeMCU microcontroller and an MQ-135 gas sensor.
- Shows readings on an LCD or web dashboard and sends data to an IoT platform for remote monitoring, storage and visualization.
- Includes threshold-based alerts for poor air quality, intended for classrooms, laboratories, homes and offices.

SKILLS

Programming: Python, Java

Web Development: HTML, CSS, JavaScript

Databases: SQL

Data Science: Data analysis, data cleaning, data visualization, basics of machine learning

Soft Skills: Analytical thinking, communication, teamwork, quick learning